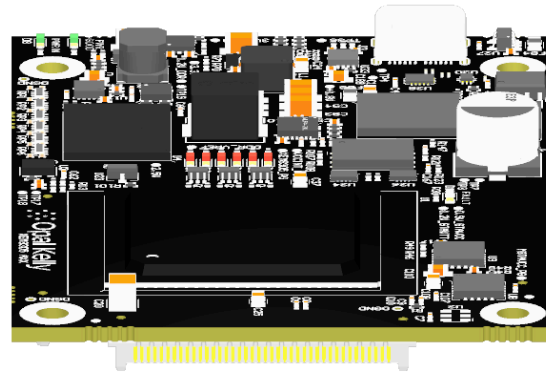
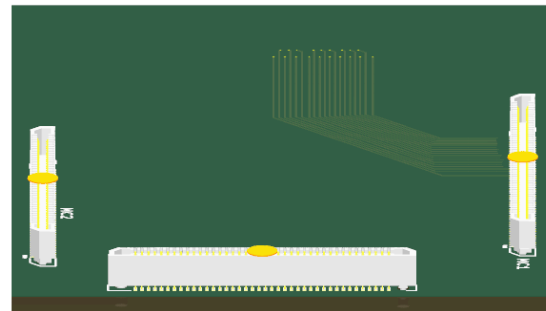


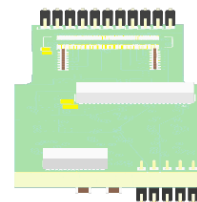
XEM8305, carrier and LDO



XEM8305



Carrier board



Current LDO

Exploded preview; open 3D for the connected view. Connector envelopes and seating are nominal.

Major updates on 10/7

1. FPGA PCB update

Material list complete. Finalizing schematic and layout; routing has not started.

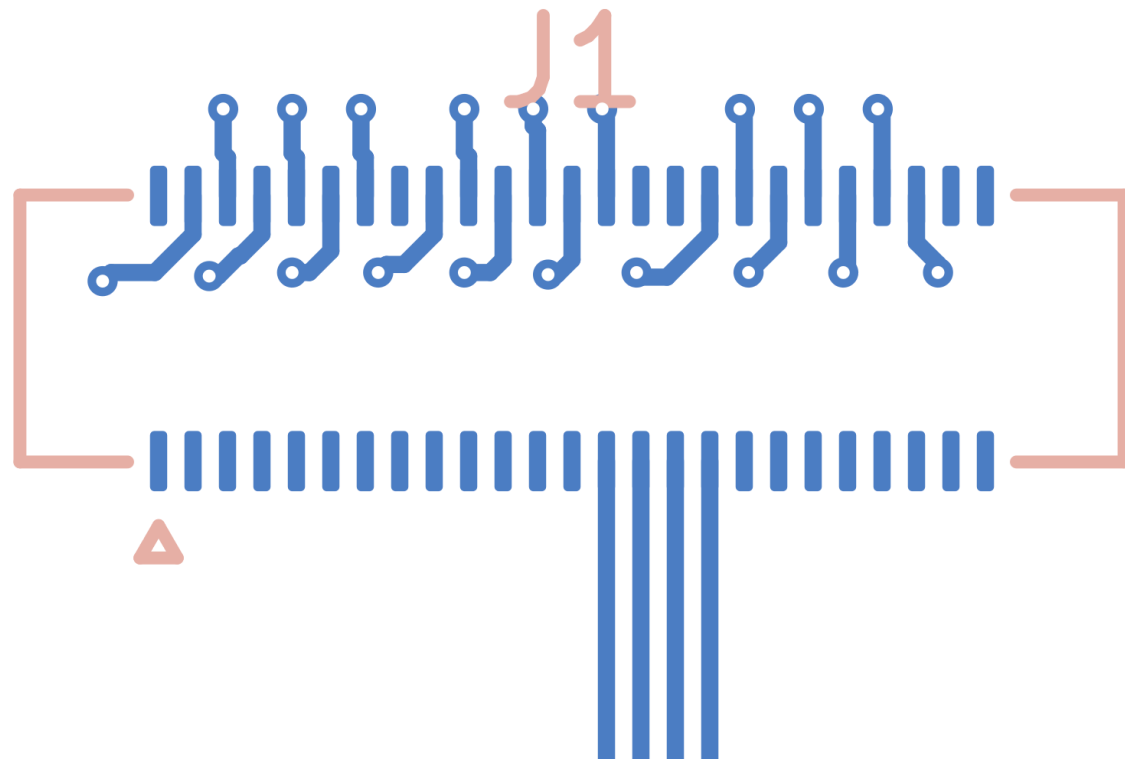
2. Alternative method: LDO → XEM8305 module

Working on schematic and routing; confirming dimensions.

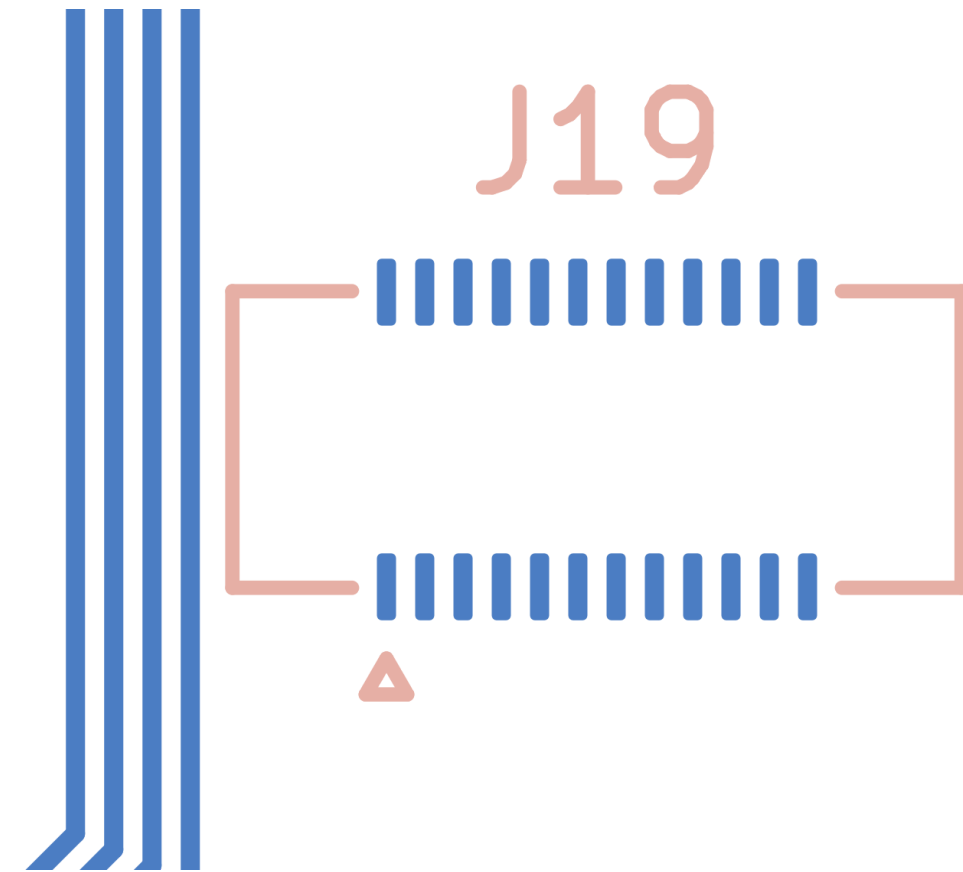
3. Moving forward

Next week: finish design review, confirm connector fit and dimensions, then begin FPGA routing.

J1 and J19 connections



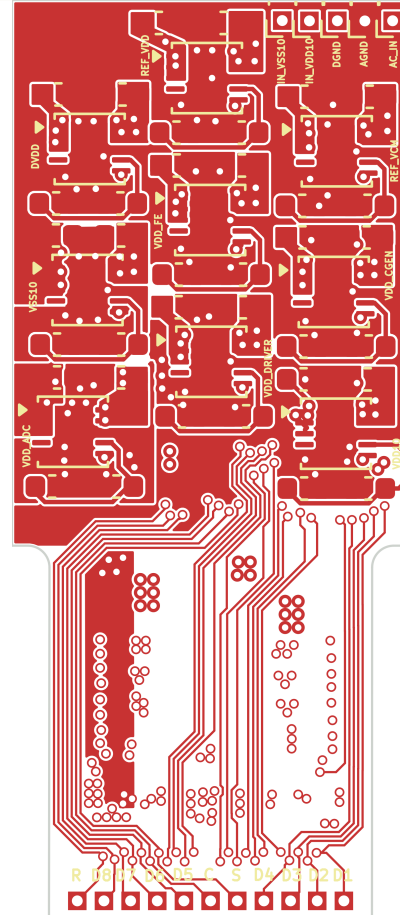
J1 · native carrier connector footprint



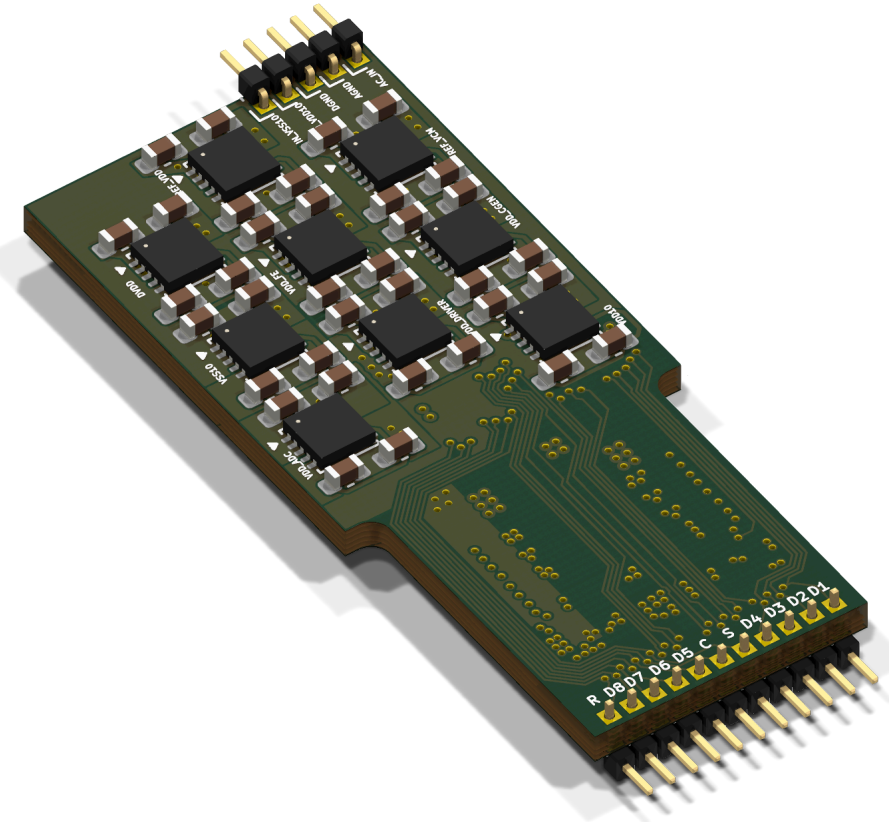
J19 · native carrier connector footprint

Carrier J1 and J19 mate with matching LDO connectors. J19 has no assigned source nets.

Existing ASIC LDO board



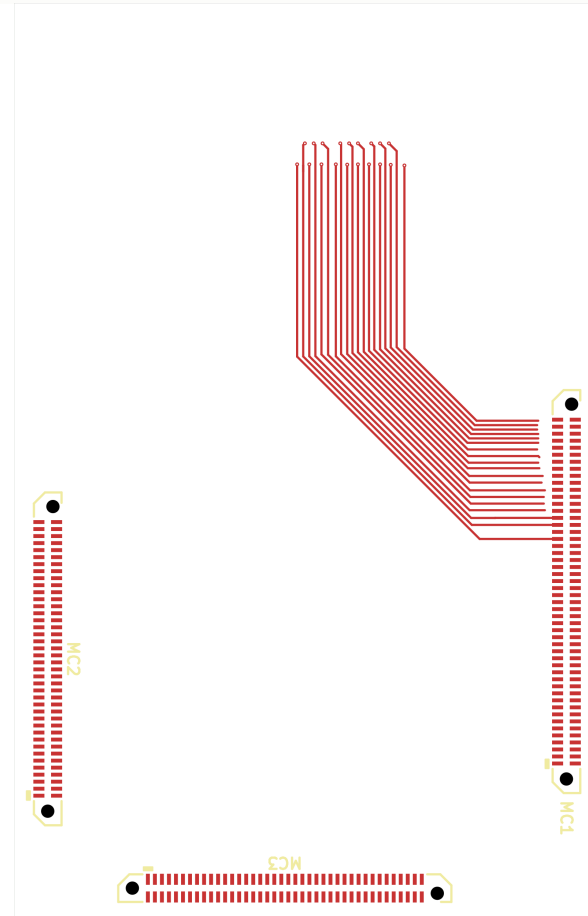
Actual native board layout



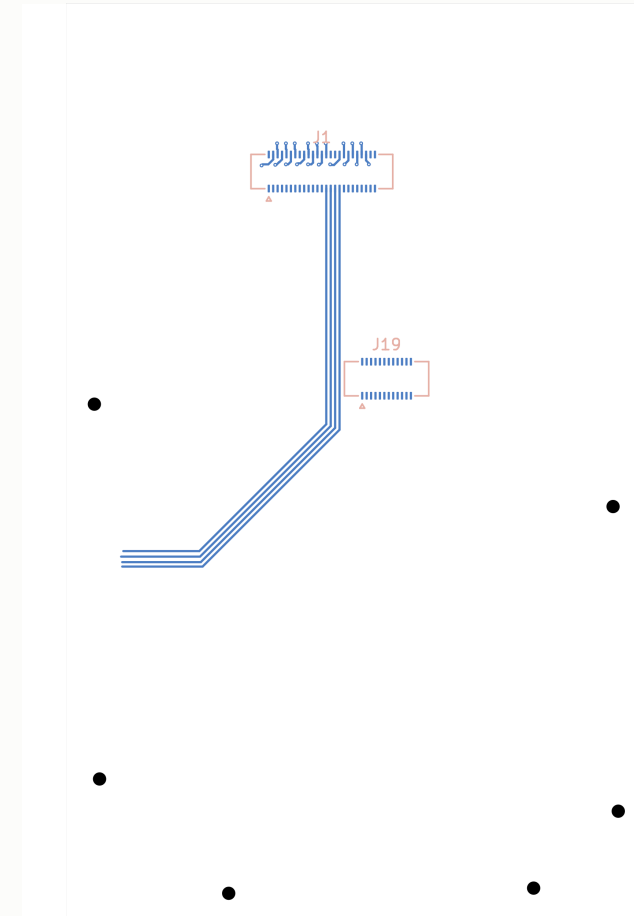
Native KiCad 3D view

Nine regulator stages · 23 direct digital paths · rail and power-order checks remain open.

Current carrier PCB



New Desktop carrier, top layout



New Desktop carrier, underside layout

Received routing draft · 20 assigned open connections · power/startup circuit pending.

Zitong's FPGA and LEDs

XC7A35T-2CSG325I: four GTP lanes and 150 user I/Os.

Eight ASICs need 117 user I/Os; four chips can stimulate.

Two LEDs: power and FPGA DONE, in different colors.

Zitong's reference clock

AX3DCF1-125.0000T3: 125 MHz, 1.8 V LVDS oscillator.

100 nF external AC-coupling capacitors.

LMK1D2102LRGTR: split the clock to MGTREFCLK and MRCC.

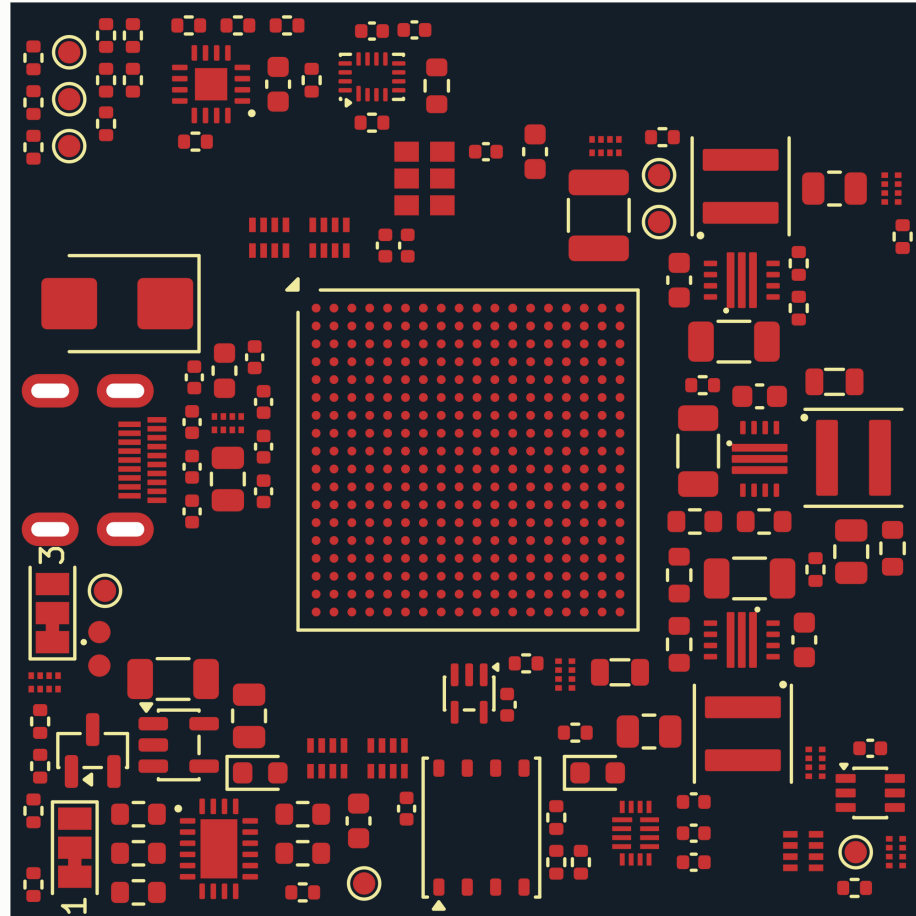
Zitong's flash and level translation

IS25WP128F-JKLE: 128 Mbit SPI NOR flash, 6 × 5 mm.

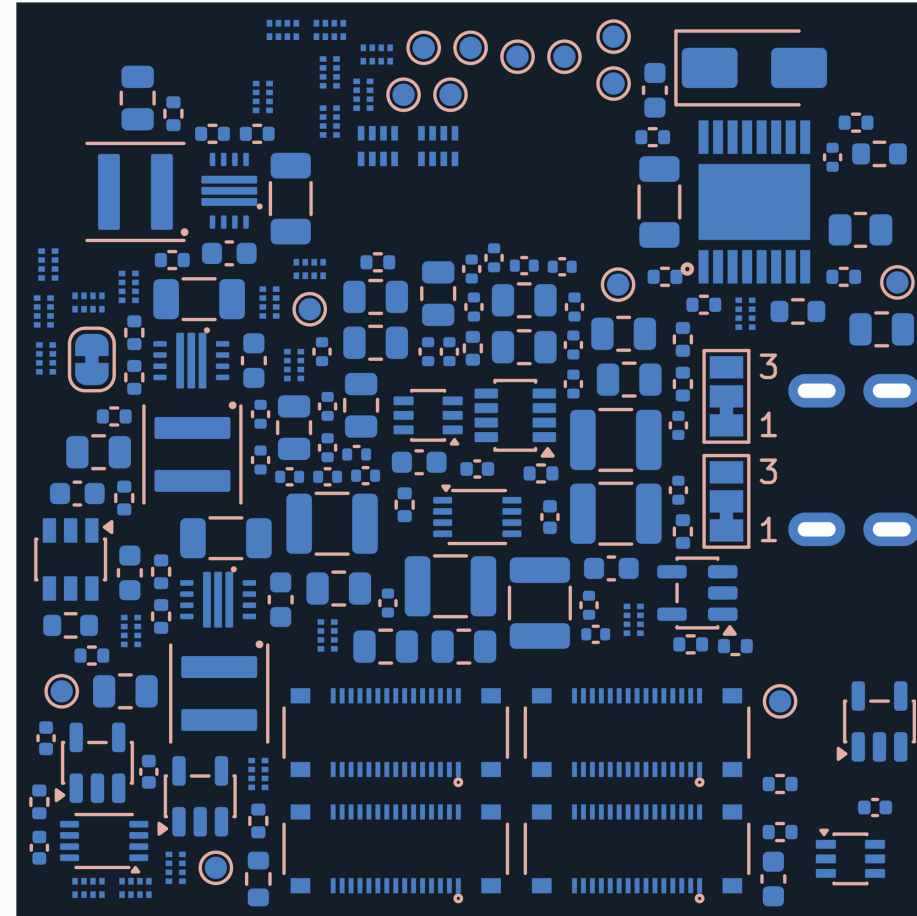
NVT4858HKZ: FPGA 1.5 V to flash 1.8 V translation.

Programming path: MCU → JTAG → FPGA → SPI → Flash.

Current FPGA PCB



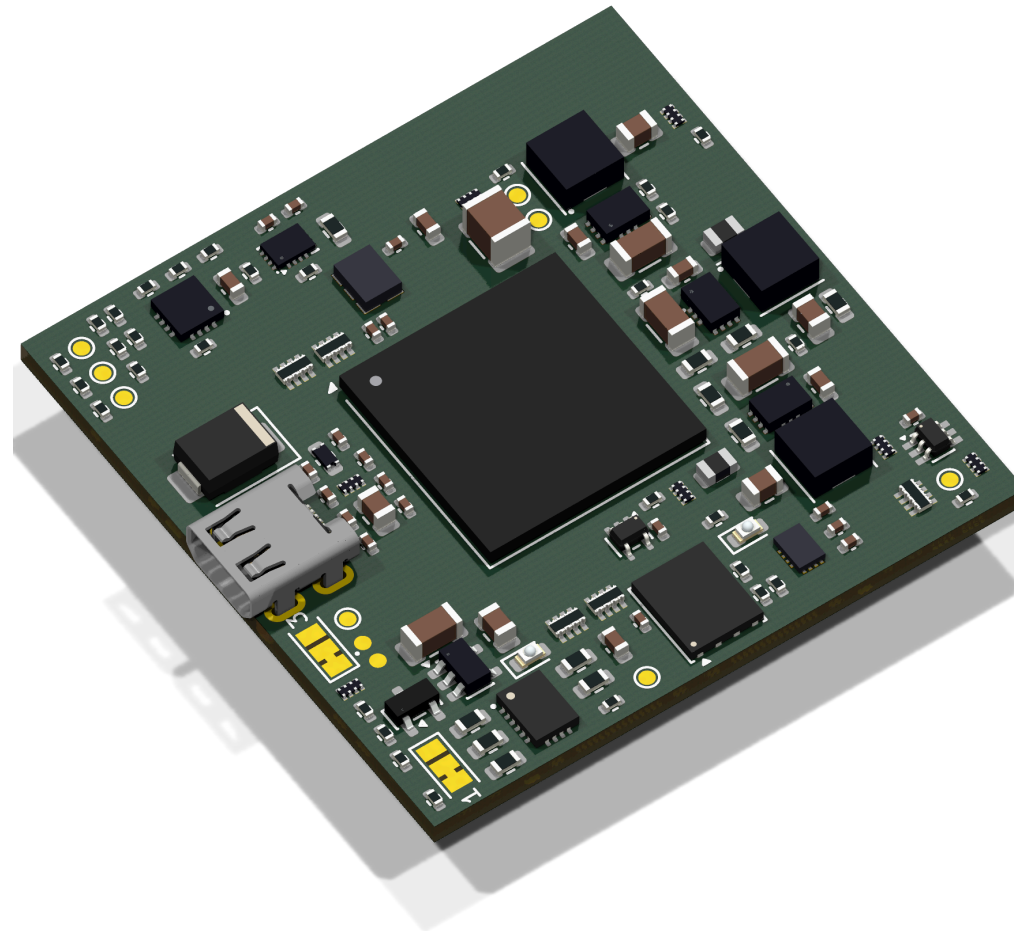
Component side



Underside · four mezzanine connectors

41 × 41 mm · 287 parts · 1,091 connections await routing.

Current FPGA board in 3D



Native KiCad model. Oscillator and translator bodies are nominal envelopes.

Gerald's board-design guidance

Keep related clock and data paths short and matched.

Place bypass capacitors close to ICs and keep continuous ground returns.

Use compact, regular placement and manufacturer-approved FPGA escapes.